



Tech to revolutionise night vision

Researchers from ANU have developed new ultra-thin tech that revolutionizes night-vision. <https://digit.in/jul21-03>



Nuclear energy by Jeff Bezos

Jeff Bezos-backed company General Fusion is building a nuclear fusion plant in the UK. <https://digit.in/jul21-04>

exponentially expanding over great distances creating tiny quantum scale fluctuations that eventually gave rise to the stars and galaxies that we see today. These patterns could inform astronomers about the processes that took place in the cosmic dark age, after the universe came into being but before there was light. Producing more sensitive CMB maps also allows astronomers to better probe the data, and run statistical studies to determine how closely the observed data matches theoretical predictions.

MONSTERS ON THE EDGES

In the latest photons to hit our astronomical instruments, the light that is only now reaching us are images of gigantic black holes, behemoths that are simply too big to exist for how old they are. These black holes at the edge of our range, seem to have consumed more matter than is possible, given the time they had. The supermassive black holes are simply too large to be allowed, which is eerie, similar to finding a healthy fish in an abandoned aquarium.

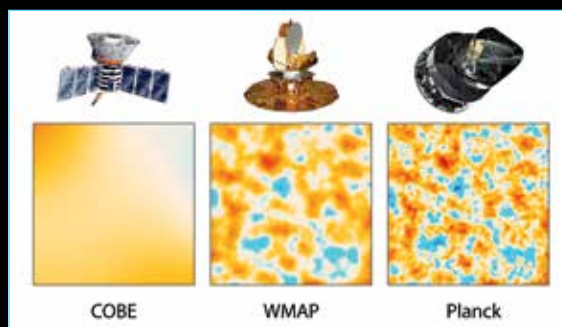
Researchers believe the discrepancy is because astronomers have so far not accounted for the dark matter in their theories and simulations. According to conventional wisdom, clouds of gas would coalesce to form the first stars and planets, then collapse to form the first seed black holes, which can be considered to be baby supermassive black holes. Adding dark matter to the clouds of material accelerates the process of collapse, allowing for far heavier seed black holes than previously believed possible, which can then devour the surrounding matter at a sufficiently high rate to explain the black holes that we are seeing from the dawn of time. The researchers found in their simulations that dark matter was the missing ingredient that neatly explained how the early black holes got so massive so quickly. Apart from providing tantalising directions for studying

the supermassive black holes in the Milky Way and surrounding galaxies, the simulations also indicate that dark matter may interact with other dark matter, providing evidence that dark matter particles do collide with other dark matter particles. The next generation of space telescopes meant for probing dark matter, such as the Giant Magellan Telescope can potentially test this new theory. The Hobby-Eberly Telescope Dark Energy Experiment (HETDEX) is an instrument designed to conduct a dark energy survey of the sky, and will also

a galaxy. While there are tantalising hints of such a signal being picked up, researchers have not been able to declare their findings with the required confidence... yet. Stochastic gravitational waves can potentially be a rich source of information about the events that took place in the young universe.

Axions are incredibly convenient at explaining a range of unexplained physics, and are believed to be the constituent particles of dark matter. Read more about Axions, fast and light particles that barely interact

with matter in the June 2021 issue of Digit. Physicists know that these particles should exist, but have not been able to spot them yet, although the hunt is going on in earnest. All the compelling theoretical reasons that point to the existence of axions also indicate that



Comparison of CMB data gathered by successive generations of satellites

investigate the large-scale structures of the universe around us, mapping around a million galaxies and looking for patterns within them.

DARK HORIZON

The farther we look, the further back in time we go. Similar to the CMB is the gravitational wave background noise from the random large scale interactions in the universe, such as galaxies colliding or merging supermassive black holes. Known as stochastic gravitational waves, astronomers are attempting to capture these elusive waves by precisely measuring millisecond pulsars, well timed beacons that fluctuate slightly when gravitational waves pass through the Earth, allowing the gravitational waves to be identified. By using other pulsars in the milky way in a similar way that a smartphone uses a GPS constellations, the approach essentially creates an astronomical instrument the size of

they must have been produced in immense quantities in the immediate aftermath of the Big Bang. The scholars reason that just as there is the CMB for photons, there must be an analogue for axions, the hypothetical cosmic axion background (CAB). The research comes about just as a number of instruments around the world are gearing up for the search for axions, and these sensitive instruments may be sensitive enough to pick up the CAB, which would be a double discovery, proving both the existence of CAB as well as axions. The researchers also suggest modifications to the existing equipment that could make them more suited for deliberately hunting for the CAB, instead of going looking for the axions. Such a discovery would help us understand some fundamental questions about the cosmos, such as if it went through a period of inflation in its history, how hot it got, and the nature of dark matter. 